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Dr. Robin Brabham
Senior Publishing Editor, Green Chemistry
Royal Society of Chemistry

Subject: Revised GC-ART-01-2024-000392.R1

Dear Dr. Brabham,

We are enclosing a revised version of our research article titled **'Integrating climate policies in the sustainability analysis of green chemicals'** to contribute to the *Green Chemistry* 25th Anniversary themed collection. We are grateful for the positive feedback from the Reviewers regarding the actions taken in response to their comments in the previous round of review.

We appreciate the acknowledgement from Reviewer #1 that we have satisfactorily addressed their concerns. Regarding Reviewer #2, we have made a concerted effort to address the Reviewer's comments and suggestions, significantly enhancing the clarity and comprehensiveness of our manuscript. Specifically, we have added clarity to the adaptation of United States gas and electricity prices for a global context, hydrogen delivery pressures, and the application of carbon capture and storage to our production routes. Additionally, we emphasise the global and regional scope of our work, ensuring consistency and clarity for our readers. We have meticulously detailed the utilisation of literature data and/or *premise*-generated processes, reinforcing transparency regarding the life cycle inventories employed in our work. Furthermore, we have addressed concerns raised by Reviewer #2 regarding the interpretation of net CO_2 capture through green methanol production routes using Direct Air Capture technology on a cradle-to-gate basis. We acknowledge the limitations of this approach and have provided a clearer discussion in the manuscript.

We are convinced that the revisions have further improved the clarity and depth of our study, trusting that they render our manuscript suitable for publication in *Green Chemistry*. Please find attached the point-by-point responses to the Reviewers' comments, the revised manuscript, and electronic supplementary information, presented with and without changes mark-up, along with the table of contents entry.

We sincerely thank you for your consideration and look forward to your response.

Best wishes,

Prof. Dr. Gonzalo Guillén-Gosálbez on behalf of all the co-authors.